

BMW Price Estimation using Gradient Boosting

Machine Learning · Regression · End-to-End Deployment

Data Science Portfolio Project · UK Used Car Market

R ² SCORE	MAE	TEST RECORDS	ALGORITHM
0.94	~£1,400	1,078	GBM

1. Problem Overview

The used car market is highly dynamic, with vehicle prices influenced by a complex interplay of depreciation, mileage, mechanical specifications, and market demand. Accurate price estimation helps both buyers and sellers make informed, data-driven decisions — reducing information asymmetry in the second-hand automotive market. This project focuses on building a robust machine learning model to estimate the resale price of used BMW vehicles from structured tabular data, covering features such as manufacturing year, mileage, engine size, fuel type, transmission type, road tax, and fuel efficiency. The dataset comprises approximately 10,000 UK BMW listings sourced from a publicly available used car dataset, representing a broad cross-section of the UK market across multiple model lines and registration years.

2. Exploratory Data Analysis

Prior to modelling, a thorough exploratory analysis was conducted to understand the structure, distribution, and relationships present in the data.

Price Distribution. Vehicle price is heavily right-skewed, with the majority of listings priced below £40,000. A small number of high-value models such as the M5, M6, and i8 create a long upper tail, indicating the presence of premium outliers that any model must handle gracefully.

Age vs. Price Relationship. A strong monotonic relationship was observed between car age and resale value — as vehicles age, their market price consistently declines. Scatter plots revealed clear non-linearity in this relationship, particularly between manufacturing year and price, confirming that simple linear assumptions would be insufficient to capture the true pricing dynamics.

Feature Correlations. Mileage showed a strong negative correlation with price, consistent with real-world depreciation patterns. Engine size exhibited a moderate positive correlation, reflecting the premium placed on higher-performance variants. Fuel type and transmission type demonstrated meaningful price separation across categories.

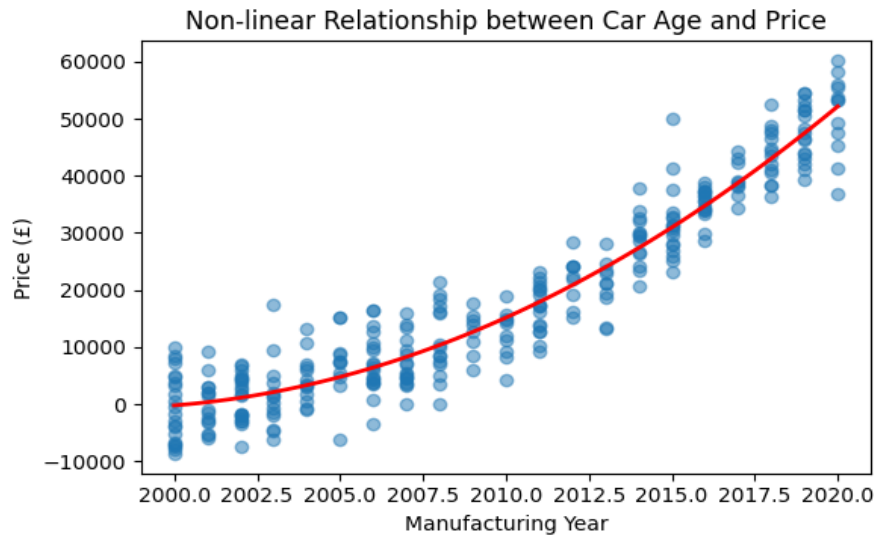


Figure 1. Non-linear relationship between manufacturing year and resale price. The polynomial trend line confirms that linear models would underfit this relationship.

3. Feature Engineering

To better capture the underlying price dynamics, domain-driven feature engineering was applied beyond the raw input variables. Three derived features were introduced:

Car Age was computed as the difference between the current calendar year and the manufacturing year. This serves as a direct proxy for depreciation and removes the implicit temporal bias present in the raw year field.

Mileage per Year was calculated by dividing total mileage by car age. This normalises usage intensity across vehicles of different ages, providing a more meaningful signal than raw mileage alone — a 5-year-old car with 100,000 km is used very differently from a 10-year-old car with the same reading.

Power Efficiency was derived as the ratio of fuel efficiency (MPG) to engine size. This composite feature captures the relationship between engine performance and economy, reflecting real-world buyer preferences for efficient yet powerful vehicles.

Categorical attributes — model name, transmission type, and fuel type — were encoded using one-hot encoding to produce binary indicator variables. Numerical features were scaled using standard normalisation to ensure balanced gradient updates during model training.

4. Model Selection Rationale

Multiple regression approaches were evaluated before arriving at the final model choice.

Linear Regression and Regularised Variants (Ridge, Lasso) achieved reasonable R^2 scores under cross-validation but consistently underperformed on absolute price accuracy. The non-linear relationships identified during EDA meant these models systematically over- or under-estimated prices at the extremes of the price distribution.

Decision Trees captured non-linearity effectively but were prone to overfitting — memorising training patterns rather than learning generalisable pricing rules. Performance on held-out data was unstable.

Gradient Boosting Regression was ultimately selected as the optimal approach. By sequentially fitting shallow trees to the residual errors of prior iterations, the algorithm effectively captures complex non-linear feature interactions while maintaining strong generalisation. Its built-in regularisation mechanisms control overfitting without requiring extensive manual tuning.

5. Model Evaluation

The final Gradient Boosting model was evaluated on a representative 10% hold-out sample of the dataset comprising 1,078 records. This evaluation strategy was chosen to ensure statistical stability and reduce variance compared to smaller test splits.

The model achieved an **R^2 score of 0.94**, indicating that 94% of the variance in used BMW prices is explained by the model's learned relationships. The **Mean Absolute Error (MAE) of approximately £1,400** confirms that individual predictions are, on average, within a realistic and commercially acceptable range for used car valuation tasks. Together, these metrics validate that the model generalises well to unseen vehicles and produces reliable, actionable price estimates.

6. Deployment

The trained model and preprocessing pipeline were serialised using Joblib and deployed as an interactive web application built with Streamlit. The application accepts user-specified vehicle parameters through an intuitive interface and returns an instant price estimate along with a confidence range. The app is publicly accessible and hosted on Streamlit Cloud, enabling real-world demonstration of the end-to-end machine learning pipeline.

7. Limitations

While the model demonstrates strong performance on the evaluation dataset, several limitations should be acknowledged. The model is trained exclusively on UK market data and may not generalise reliably to other regional markets where pricing dynamics differ. The dataset reflects a specific historical time window and does not account for macroeconomic shifts such as the post-2020 used car price surge driven by semiconductor shortages. Additionally, condition-related factors such as service history, accident records, and cosmetic state — which significantly influence real-world valuations — are not captured in the available features.

8. Conclusion

This project demonstrates the importance of aligning model selection with data characteristics and practical business objectives. By prioritising absolute prediction error alongside statistical fit metrics, and by leveraging Gradient Boosting's capacity to model complex non-linear interactions, the solution delivers practical, portfolio-ready results suitable for real-world deployment. The end-to-end pipeline — from raw data through EDA, feature engineering, model training, evaluation, and live web deployment — reflects a complete applied machine learning workflow.

Built with Python · Scikit-learn · Streamlit · UK BMW Dataset | Predictions are estimates for informational purposes only.